

# Quantum field theory cheat sheet

## – path-integral formulation –

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### Abstract

From Peskin Schroeder “An introduction to Quantum Field Theory Chapter 9”

| Green function in path integral formulation           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| (P) $n$ -point Green function or correlation function | <p>The probability amplitude to observe <math>k_1, k_2, \dots, k_n</math> particles described by a field <math>\phi</math> at space-time positions <math>x_1, x_2, \dots, x_n</math> is given by the <b><math>n</math>-point Green function</b> or <b><math>n</math>-point correlation function</b> denoted as <math>\langle \Omega   T [\phi_H^{k_1}(x_1) \phi_H^{k_2}(x_2) \cdots \phi_H^{k_n}(x_n)]   \Omega \rangle</math>.</p> <p>According to the path-integral formulation, this function can be obtained from a Lorentz invariant Lagrangian <math>\mathcal{L}</math>:</p> $\frac{\int \mathcal{D}\phi \phi^{k_1}(x_1) \phi^{k_2}(x_2) \cdots \phi^{k_n}(x_n) \exp \left[ i \int d^4x \mathcal{L}(\phi, \partial_\mu \phi) \right]}{\int \mathcal{D}\phi \exp \left[ i \int d^4x \mathcal{L}(\phi, \partial_\mu \phi) \right]}$ <p>Note that in the above formula, there are no operators involved.</p> <p>Note that each degree of freedom of <math>\phi</math> (as a real number, i.e. a complex field has 2 degrees of freedom) represents a degree of freedom for the particle (spin, particle/anti-particle).</p> <p>Note that the observed fields must be explicitly listed. For instance, absence of <math>\phi^{k_2}(x_2)</math> above means that there is no particle at point <math>x_2</math>, which is not equivalent to saying that we do not observe what happens at <math>x_2</math>.</p> |

| <b>Functional derivatives, generating function</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| (D) <b>Functional derivatives for fields</b>       | <p>If space-time was discretized and each space-time point indexed by <math>i</math>, a field would be a vector <math>(\phi_i)_i</math> and if we have a functional on the field <math>\phi</math>, a derivative with respect to <math>\phi_i</math> for some space-time point indexed by <math>i</math> would be defined as a standard derivative with respect to <math>\phi_i</math>. Namely <math>\frac{\partial}{\partial x_i} x_j = \delta_{ij}</math> and <math>\frac{\partial}{\partial x_i} \sum_j x_j k_j = k_j</math>.</p> <p>The <b>functional derivative</b> is an extension of this for fields defined on a continuous space-time:</p> $\frac{\delta}{\delta J(x)} J(y) = \delta(x - y), \quad \text{and} \quad \frac{\delta}{\delta J(x)} \int d^4 y J(y) \psi(y) = \psi(x)$ <p>For composite functions, the ordinary differentiation rules applies:</p> $\frac{\delta}{\delta J(x)} f(G[J]) = \frac{\delta G[J]}{\delta J(x)} f'[G[J]], \text{ for instance}$ $\frac{\delta}{\delta J(x)} \exp \left[ i \int d^4 y J(y) \psi(y) \right] = i \psi(x) \exp \left[ i \int d^4 y J(y) \psi(y) \right]$ <p>For functional on derivatives of <math>J</math>, first integrate by parts to remove the derivatives (assuming surface terms are zero), for instance:</p> $\frac{\delta}{\delta J(x)} \int d^4 y (\partial_\mu J(y)) V^\mu(y) = -\partial_\mu V^\mu$ |
| (D) <b>Generating functional</b>                   | <p>The correlation functions can be written as functional derivatives of a functional <math>Z[J]</math> called <b>generating functional</b>:</p> $Z[J] = \int \mathcal{D}\phi \exp \left[ i \int d^4 x [\mathcal{L}(x) + J(x)\phi(x)] \right]$ <p>for instance</p> $\langle \Omega   T [\phi_H(x_1) \phi_H(x_2)]   \Omega \rangle = \frac{1}{Z_0} \left( -i \frac{\delta}{\delta J(x_1)} \right) \left( -i \frac{\delta}{\delta J(x_2)} \right) Z[J] \Big _{J=0}$ <p>where <math>Z_0 = Z[J = 0]</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| <b>Free real scalar fields</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| (P) <b>Generating functional</b> | <p>The free Lagrangian of the real scalar field is</p> $\mathcal{L}_0 = \frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2$ <p>The exponent for the generating functional reads, after an integration by part and adding a <math>i\epsilon</math> component so that the integral converges:</p> $\int d^4 x [\mathcal{L}_0 + J\phi] = \int d^4 x \left[ \frac{1}{2} \phi (-\partial^2 - m^2 + i\epsilon) \phi + J\phi \right]$ <p>using the change of variable <math>\phi(x) = \phi'(x) + i \int d^4 y \Delta_F(x - y) J(y)</math> with Jacobian 1 (where <math>\Delta_F</math> is the Feynman propagator), using a double integration by parts and using the fact that <math>\Delta_F(x - y)</math> is a Green function for the operator <math>(\partial^2 + m^2)</math> we get the generating functional:</p> $Z[J] = Z_0 \exp \left[ -\frac{1}{2} \int d^4 x d^4 y J(x) \Delta_F(x - y) J(y) \right]$ <p>we check in particular that</p> $\langle 0   T [\phi(x_1) \phi(x_2)]   0 \rangle = -\frac{\delta^2}{\delta J(x_1) \delta J(x_2)} Z[J] = \Delta_F(x_1 - x_2)$ |

| Free electromagnetic fields |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| (P) Generating functional   | <p>The free Lagrangian for the electromagnetic field is <math>\mathcal{L}_0 = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu}</math>, where <math>F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu</math></p> <p>To deal with the gauge invariance, i.e. invariance under the shift <math>A_\mu(x) \mapsto A_\mu(x) + \frac{1}{e}\partial_\mu\alpha(x)</math> we restrict the functional integration to the field <math>A_\mu</math> that nearly obeys the Lorentz gauge <math>G_\omega(A) = \partial^\mu A_\mu(x) - \omega(x) = 0</math> with <math>\omega</math> small. In practice we integrate over the functions <math>\alpha</math> such that <math>G(A) = 0</math> and integrate over the function <math>\omega</math> with the weight <math>\exp\left[-i\int d^4x \frac{\omega^2}{2\xi}\right]</math>. This gives, noting that the action <math>S[A]</math> and any observable <math>O(A)</math> being gauge invariant:</p> $\begin{aligned} & \int \mathcal{D}A e^{iS[A]} \\ &= \int \mathcal{D}\omega e^{-i\int d^4x \frac{\omega^2}{2\xi}} \det(G_\omega) \int \mathcal{D}\alpha \int \mathcal{D}A e^{iS[A]} \delta(G_\omega(A)) \\ &= \det(G_\omega) \int \mathcal{D}\alpha \int \mathcal{D}A e^{iS[A]} \exp\left[-i\int d^4x \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right] \end{aligned}$ <p>where <math>\det(G_\omega) = \det(\partial^2/e)</math> is independent of <math>\alpha</math> and <math>A</math>. We get:</p> $\langle \Omega   T[O(A)]   \Omega \rangle = \frac{\int \mathcal{D}A O(A) \exp\left[i\int d^4x \left[\mathcal{L}_0 - \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right]\right]}{\int \mathcal{D}A \exp\left[i\int d^4x \left[\mathcal{L}_0 - \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right]\right]}$ |
| Grassmann numbers           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| (D) Grassmann numbers       | <p>By definition, if <math>\theta</math> and <math>\eta</math> are <b>Grassmann numbers</b> and <math>a</math> an ordinary number</p> <p><b>(anticommutation)</b> <math>\theta\eta = -\eta\theta</math> thus <math>\theta^2 = 0</math></p> <p><b>(integral, real number)</b> <math>\int d\theta a = 0</math>, <math>\int d\theta \theta = 1</math></p> <p><b>(integral, complex number)</b> <math>\theta</math> and <math>\theta^*</math> are considered as independent numbers, and <math>\int d\theta^* d\theta (\theta\theta^*) = 1</math></p> <p><b>(derivative)</b> <math>\frac{\partial}{\partial\eta}\theta\eta = -\frac{\partial}{\partial\eta}\eta\theta = -\theta</math></p> <p>Addition, multiplication with ordinary numbers follow ordinary rules, in particular <math>a\theta = \theta a</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (P) Properties              | <p>For any Grassman numbers <math>\theta</math>, and ordinary numbers <math>b</math>, and Hermitian matrix <math>B</math>:</p> <p>(1) any function of <math>\theta</math> reads <math>f(\theta) = a + b\theta</math> (Taylor expansion)</p> <p>(2) in an integration <math>\int d\theta_1 d\theta_2 \dots</math> only the integrand <math>\theta_1\theta_2 \dots</math> contributes</p> <p>(3) <math>\int d\theta^* d\theta e^{-\theta^* b \theta} = b</math></p> <p>(4) <math>\int d\theta^* d\theta \theta \theta^* e^{-\theta^* b \theta} = 1</math></p> <p>(5) <math>(\prod_i \int d\theta_i^* d\theta_i) e^{-\theta_i B_{ij} \theta_j} = \det B</math></p> <p>(6) <math>(\prod_i \int d\theta_i^* d\theta_i) \theta_k \theta_l e^{-\theta_i B_{ij} \theta_j} = \det B (D^{-1})_{kl}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| <b>Free spinor fields</b>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| (P) Generating functional      | <p>The free Lagrangian for the spinor field is</p> $\mathcal{L}_0 = \bar{\psi}(i\cancel{\partial} - m)\psi$ <p>The spinor field is a Grassmann field i.e. it reads as</p> $\psi(x) = \sum_i \psi_i \phi_i(x)$ <p>where <math>\phi_i(x)</math> is the basis of four-component spinors made of ordinary complex numbers and <math>\psi_i</math> are constant Grassmann complex numbers.</p> <p>The generating functional reads</p> $Z[\bar{\eta}, \eta] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[ i \int d^4x \left[ \bar{\psi}(i\cancel{\partial} - m)\psi + \bar{\eta}\psi + \bar{\psi}\eta \right] \right]$ $= Z_0 \exp \left[ - \int d^4x d^4y \bar{\eta}(x) S_F(x - y) \eta(y) \right]$ |
| <b>Quantum electrodynamics</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (P) Generating functional      | <p>The full Lagrangian reads</p> $\mathcal{L} = \bar{\psi}(i\cancel{\partial} - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e \bar{\psi} \gamma^\mu A_\mu \psi$ $= \mathcal{L}_0 - e \bar{\psi} \gamma^\mu A_\mu \psi$ <p>The generating function for the full theory can be written perturbatively expanding the exponential for the interaction term:</p> $\exp \left[ i \int d^4x \mathcal{L} \right] = \exp \left[ i \int d^4x \mathcal{L}_0 \right] \times \left\{ 1 - ie \int d^4x \bar{\psi} \gamma^\mu A_\mu \psi + \dots \right\}$                                                                                                                                                              |